

3. PREPARE TEST PLAN

A Test Plan is a formal document that defines the testing strategy, objectives, scope, resources, schedule, and deliverables for a software testing project. It serves as a roadmap for all testing activities and ensures that testing is performed in a structured and organized manner.

For the **Web Application Security Testing of DVWA using OWASP ZAP**, the Test Plan outlines how functional and security testing activities will be performed, which tools will be used, and the criteria for starting and completing testing.

3.1 Test Strategy

The Test Strategy defines the overall testing methodology that will be followed throughout the project.

The primary objective of this project is to verify the security of the DVWA application by executing automated test cases and vulnerability scans. The strategy focuses on identifying security weaknesses and validating application behavior under both normal and malicious conditions.

The testing strategy includes:

Functional Testing

Functional testing is performed to verify that the application behaves as expected during normal user interactions.

The following functionality is tested:

- User Login
- Navigation between modules
- Form submissions
- User authentication

Security Testing

Security testing is the primary focus of this project.

The following vulnerabilities are tested:

- SQL Injection (SQLi)
- Cross-Site Scripting (XSS)
- Security Misconfigurations
- Missing Security Headers

Automated Testing

Automation is implemented to improve efficiency, consistency, and repeatability.

Automation tools used include:

- Selenium WebDriver
- TestNG Framework
- Maven Build Tool
- OWASP ZAP

Vulnerability Assessment

OWASP ZAP is integrated with the automation framework to:

- Discover application pages
- Identify vulnerabilities
- Classify security risks
- Generate vulnerability reports

Reporting Strategy

Multiple reporting mechanisms are used to capture execution results and findings.

Reports generated include:

- TestNG Reports
- Extent Reports
- OWASP ZAP Security Reports

The strategy ensures complete visibility into application behavior and security vulnerabilities.

3.2 Test Approach

The Test Approach describes how testing activities will be executed to achieve project objectives.

The project follows a combination of Functional Testing and Security Testing approaches.

Step 1: Environment Preparation

Before testing begins, the required environment is configured.

Activities include:

- Installing Java 21

- Configuring Eclipse IDE
- Installing Maven
- Installing XAMPP
- Setting up DVWA
- Configuring MySQL Database
- Installing OWASP ZAP

Step 2: Framework Development

The automation framework is developed using:

- Java Programming Language
- Selenium WebDriver
- TestNG Framework
- Maven Dependency Management

The framework follows a modular structure to improve maintainability and reusability.

Step 3: Test Case Design

Test cases are designed for:

- Login Validation
- SQL Injection Testing
- Cross-Site Scripting Testing
- OWASP ZAP Security Scans
- Reporting Validation

Step 4: Functional Test Execution

Login functionality is executed first to ensure application accessibility.

The following are validated:

- Successful login
- Navigation behavior
- Authentication process

Step 5: Security Test Execution

Security tests are executed using malicious payloads.

Testing includes:

- SQL Injection payload execution
- XSS payload execution
- Input validation testing

Step 6: OWASP ZAP Scanning

OWASP ZAP is used to perform:

Spider Scan

The Spider Scan automatically crawls the application and identifies available pages and endpoints.

Active Scan

The Active Scan tests discovered pages for known security vulnerabilities.

Step 7: Result Analysis

After execution, all results are analyzed.

Analysis includes:

- Passed test cases
- Failed test cases
- Security vulnerabilities
- Risk severity levels

Step 8: Report Generation

Reports are generated and reviewed.

The reports include:

- Execution summaries
- Vulnerability findings
- Risk classifications
- Remediation recommendations

3.3 Entry Criteria

Entry Criteria define the conditions that must be satisfied before testing activities can begin.

The following conditions must be met before test execution:

Environment Availability

The testing environment must be fully configured and operational.

Required components:

- Java 21 installed

- Eclipse IDE configured
- Maven configured
- XAMPP running
- Apache Server active
- MySQL Database active
- DVWA application accessible
- OWASP ZAP installed and running

Test Data Availability

Required test data must be prepared.

Examples include:

- Valid login credentials
- SQL Injection payloads
- XSS payloads

Framework Readiness

The automation framework must be ready for execution.

Requirements:

- Selenium scripts developed
- TestNG configured
- Maven dependencies resolved

Tool Verification

Testing tools must be functioning correctly.

Tools include:

- Chrome Browser
- ChromeDriver
- Selenium WebDriver
- OWASP ZAP

Test Cases Prepared

All test cases and scenarios must be reviewed and approved before execution.

3.4 Exit Criteria

Exit Criteria define the conditions that must be satisfied before testing activities can be considered complete.

Testing will be considered complete when:

Test Case Execution Completed

All planned test cases have been executed.

This includes:

- Login Test Cases
- SQL Injection Test Cases
- XSS Test Cases
- OWASP ZAP Test Cases
- Reporting Test Cases

Security Scans Completed

OWASP ZAP scans have successfully completed.

Required scans:

- Spider Scan
- Active Scan

Reports Generated

All required reports have been generated.

Reports include:

- TestNG Report
- Extent Report
- OWASP ZAP Security Report

Findings Documented

All vulnerabilities and observations have been documented.

Documentation includes:

- Vulnerability description
- Severity level
- Impact analysis
- Recommendations

Critical Issues Reviewed

Any critical vulnerabilities discovered during testing must be reviewed and analyzed.

Project Documentation Completed

Final project documentation and reports must be completed and reviewed.

3.5 Test Deliverables

Test Deliverables are the outputs produced throughout the testing lifecycle.

The following deliverables are included in this project:

Test Plan Document

Contains:

- Testing objectives
- Scope
- Strategy
- Schedule
- Resources

Test Scenarios

High-level testing conditions prepared for validation.

Test Cases

Detailed test steps and expected results for each scenario.

Test Execution Report

Contains:

- Executed test cases
- Pass/Fail status
- Execution summary

Bug Reports

Contains:

- Defect description
- Reproduction steps
- Expected result
- Actual result

- Severity and priority

TestNG Report

Generated automatically after execution.

Contains:

- Test execution statistics
- Pass/Fail summary

Extent Report

Provides detailed execution logs and screenshots.

OWASP ZAP Security Report

Contains:

- Vulnerability details
- Risk classifications
- Security findings

Final Project Report

Includes:

- Project overview
- Methodology
- Results
- Findings
- Recommendations

Source Code Repository

Contains:

- Automation scripts
- Configuration files
- Maven project structure

3.6 Resource Planning

Resource Planning identifies the hardware, software, tools, and personnel required for project execution.

Hardware Requirements

The following hardware is required:

System Configuration

- Processor: Intel i5 or higher
 - RAM: Minimum 8 GB
 - Storage: Minimum 20 GB Free Space
 - Internet Connection: Required for tool downloads
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Software Requirements

Operating System

- Windows 10 / Windows 11

Development Tools

- Java 21
- Eclipse IDE
- Maven

Testing Tools

- Selenium WebDriver
- TestNG
- OWASP ZAP

Web Environment

- XAMPP
- Apache Server
- MySQL Database
- DVWA Application

Browser

- Google Chrome
 - ChromeDriver
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Human Resources

The project can be executed by:

QA Automation Engineer

Responsibilities:

- Framework development
- Script creation
- Test execution
- Report generation

Security Tester

Responsibilities:

- Vulnerability analysis
- Security scan execution
- Risk assessment

Project Reviewer

Responsibilities:

- Documentation review
- Result validation
- Final approval

Time Planning

Activity	Estimated Duration
Requirement Analysis	1 Day
Environment Setup	1 Day
Framework Development	2 Days
Test Case Design	1 Day
Test Execution	1 Day
Security Scanning	1 Day
Report Generation	1 Day
Documentation	2 Days

Total Estimated Duration: 10 Days

This Test Plan provides a structured roadmap for executing the DVWA Security Testing Automation Project and ensures that all functional, security, reporting, and documentation activities are performed systematically and effectively.